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COTP: Coplanning and Coexecuting AIGC Agent Native Product for Predictive Maintenance and Operational Optimisation of Cross Border Transition Pipelines

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Abstract

The Cross-border long-distance gas pipeline (CLGP) is a key energy corridor for China, ensuring energy supply and optimising consumption. With improved operational precision, allied decision-making is needed for pipeline health and gas transmission planning, integrating operational status, macroeconomic factors, geopolitics, and market forecasts. This manuscript introduces pipeline decision support system product (COTP), a generative AI agent based decision support system, enabling natural language interaction, flow transmitting simulation, and predictive maintenance for CLGP.

COTP leverages multi-role generative agent technology to address gas transmission volume, pipeline operation, maintenance strategy decisions in CLGP. It operates through an autonomous loop of issue identification, code generation and work execution, output ranking, insight generation, and refinement. COTP relies entirely on natural language interaction, enabling users to initiate decision-making via prompt engineering, by agents outputting gas transmission plans, and with respect to corresponding maintenance suggestions if necessary. To enhance prediction and decision-making accuracy, COTP supports collaborative optimisation using voting and game theory achieved by multiple agents. It integrates multi-source heterogeneous data from internal and external sources via large models' online query capabilities, facilitating task breakdown and execution through deep thinking. A human-in-the-loop (HIL) design allows real-time user interaction during agent thinking and execution, ensuring decision-making aligns with user intent.

COTP's steerability, user experience, prediction accuracy, and interpretability were validated using actual pipeline operating data, including historical CLGP scheduling reports, SCADA/PPS monitoring data, inventory levels, weather conditions, and commodity price fluctuations. 12 pipeline scheduling employers (6 online, 6 offline; 6 males, 6 females) evaluated COTP by comparing it with baseline methods—direct prompt-based generative AI and optimising computation for gas transmission decisions and maintenance strategies. Volunteers provided subjective and objective ratings on a 0-5 scale (strongly disagree to completely agree) to assess steerability, user experience, problem-solving, and refinements in terms of comparative studies over the task executed by COTP and baseline, respectively.

All volunteers acknowledged COTP's steerability and user experiences, favouring its HIL, step-in interaction design over the baseline for enhanced stability. COTP demonstrated superior autonomous problem detection and robustness. Most volunteers agreed COTP enabled effective data-driven predictions and insights, while the baseline exhibited inconsistent performance and unreliable refinement ability. Although code generation category remain similar, it remained few existent since the baseline's overall process lacked significance.

This findings verified that through innovative product interaction design, COTP has achieved fully natural language prompt driven co-planning, co-executing and decision-making processes, enabling autonomous execution and refinement. It provided simpler and efficient user-experience for complex decision-making in CLGP transition operation. COTP has distilled a more agile product interaction, leveraging rapid iteration capabilities of generative AI. By adapting to different target scenarios, it has potential to achieve rapid applications in equipment management, preventive maintenance, and spare parts strategy formulation across other domains. Compared to baseline, COTP's autonomous refinement capabilities offer more reasonable, feasible, and scientifically grounded solutions for complex decision-making challenges.

Keywords: Co-planning, Co-executing, Generative AI agents, Scheduling optimisation, Cross-border pipelines

Introduction

The CLGP operated by China National Petroleum Corporation (CNPC), which lies along the "Belt and Road" initiative, spans cross 6+ countries, with a total mileage exceeding 11,000 km and oil and gas (O&G) transmission capacity of over 100 million tons per year, which is accounting for 25% of CNPC's overseas long-distance pipelines. The scientific, efficiency and reliability of CLGP scheduling now directly impact China's energy consumption. Given the specifications of CLGP, which is influenced by multiple factors such as spanning multiple countries, long distances, a mixed-ownership equity structure, as well as a multilingual and multicultural working environments, the scheduling process of CLGP is subject to the combined effects of various factors, including political factors in the host countries, commodity prices, environmental and meteorological conditions, domestic energy consumption, and dynamic oil and gas inventories (Zhou et al. 2025). Despite significant improvements in equipment reliability, scheduling standardisation, the maturity of scheduling strategies, and the standardisation of operation and maintenance through multiple rounds of automation upgrades, there is still considerable room for enhancement in scheduling efficiency, process automation and intelligence, as well as refined decision-making (Kang et al. 2024). The main constraints include the long training period for professional scheduling personnel (which might takes 1–1.5 years to train an experienced scheduler), significant efficiency room could be squeezed during scheduling processes (with over 2,000 scheduling events handled daily, with hours of time costs, and approximately 5%–8% error), lagging technology in emergency warning and response for sudden events (with a response time of over 3 hours and delays exceeding 30% in winter), and low decision supporting efficiency (traditional simulation verification accounts for over 40% of the time, and there are issues with decision steerability, etc.).

At current state-of-art, the allied scheduling of CLGP primarily relies on experience-driven scheduling strategies, combined with methods based on mathematical statistics and machine learning techniques, following an in-event emergency response + post-event reflection model. This model faces bottlenecks such as a high dependence on expert experience, a lack of event data driven collaborative simulation based decision supporting capabilities, inadequate comprehensive explanations for ambiguous forecast results, and an inability to conduct near real time simulation deductions as well as providing rapid decision recommendations based on event triggers (Alpar et al. 2024). These bottlenecks leave an open door for improvement in the timeliness, accuracy, and human-machine friendliness of key decisions that ensure the

safe operation of CLGP, such as scheduling strategies, operation and maintenance policies, supply assurance plans, and emergency scheduling decisions under the sudden events.

Thanks to the evolutionary impact of generative AI agents on key human-machine interaction elements, including dialogue driven deep thinking (Deepseek. 2025), content generation (OpenAI. 2025), professional knowledge retrieval (Edge, D et al. 2024), complex logical thinking and tool-using (Schmidgall et al. 2025; Krishnan, 2025), task execution (Feng et al. 2025), as well as reflection and optimisation (Ocker et al. 2025), it has become possible to achieve explainable human-machine interactive based decision-supporting in near real time scenarios. Given that CLGP allied scheduling decisions represent a typical human-agent-machine collaborative interaction application (Yamada et al. 2025), explainable auxiliary decision-making can be achieved through HIL approach (Ren et al. 2025). The essence of this approach is to predict changes in pipeline operation data over future time periods through time series data prediction under various conditions, and based on these predictions, enable agents to collaborate with end-users in thinking and generate insights for different prediction results, thereby achieving user-assisted decision-making from an explainable perspective. If generative agents can be applied to assist in analysing and generating operational strategies for the pipeline running system under various disturbances, such as equipment operating status, macro and micro environmental factors, seasonal factors, and extreme/unexpect conditions, it will significantly improve the operational decision-making performance, precision, and product added values of CLGP.

To address the above mentioned complex decision based issues of CLGP mentioned above, thereby leveraging even better performance by taking into account generative agents in thinking and reasoning, the manuscript present design and implementation of a novel CLGP co-planning and co-executing product (COTP) based on the fusion of multi-role genAI agent clusters and CLGP multi-source heterogeneous data, with a set of maintenance, prediction and simulation algorithms hidden behind. COTP supports natural language based human-machine interaction to construct prompt engineering, enabling a human-agent-interaction-algorithm (HAA) collaborative decision-making workflow with HIL interactions. Through innovative human-machine interaction design, it achieves allied simulation regarding to a combination of knowledge engineering, system simulation, and multi-source heterogeneous pipeline operation data, providing explainable insights for pipeline transmission volume expectations, equipment health status, energy consumption expectations, and various uncertainties, thereby supporting pipeline scheduling decision-makers in achieving allied decision-making for multiple CLGPs to reach a global optimal. The manuscript is organised as follows: section 2 describes the key algorithms of COTP design by using mathematical expression, section 3 introduces the application architecture of COTP and the motivations behind its architectural design, section 4 presents the human-machine interaction design of COTP in collaborative planning and decision-making, the verification experiments and conclusions are given in sections 5 and 6, respectively.



Figure 1—Cross-border O&G pipelines with total distance over 11,000+ km (with name desensitised)

Key algorithm for running COTP

COTP leverages multi-role generative agent technology to address gas transmission volume, pipeline operation, maintenance strategy decisions in CLGP. It operates through an autonomous loop of issue identification, code generation, output ranking, insight generation, and continuous refinement. COTP relies entirely on natural language interaction, enabling users to initiate decision-making via prompt engineering, with agents outputting gas transmission plans and corresponding maintenance suggestions. Thank to the autonomous thinking, processing and decision making of gen-AI agents, COTP proposes a novel human-machine interacting, namely, swarm genAI-agents (SGH) with HIL and make it as a standard product, to achieve human-agent-pipelines as a new type of intelligent system.

In order to stimulate full potentials regarding to the synergy of AIGC, generative agents as well as CLGP scheduling, simulating and decision supporting approaches, complex CLGP scheduling and multiple criterion decision-supporting based simulation are considered as multi-objectives optimisation by using an adaptive formed swarm agent clusters with distributed thought-chain close loop to cooperate with various AIGC models, knowledge and scientific algorithms which were pre-defined; To enumerate all possible decision supporting materials within the main thinking mind keep continuing, a side chain interaction was designed by splitting agent sub-cluster with a variety of foundation model. To guarantee the prediction and decision-making through accurate way as much as possible, COTP supports collaborative optimisation based on voting and game theory liked mechanism among multiple agents, thereby triggering autonomous multiple professional models and expert knowledge coordination as well as recommending and ranking the decision support insight with reasonable insight. The key algorithm for executing COTP is shown as follows:

Algorithm 1—Swarm genAI-agents with human-in-loop (SGH)

1. Multi-objective optimisation for transmission decision

The cross disciplinary decision supporting of CLGP is actually considered as multiple CLGP cooperative decision making based on multiple criterion and interference. In mathematical expression, let X be the set of decision variables related to gas transmission, including variable dimensionality of inventory, consumption timing, and gas flow rates. The objective is to find the optimal solution $x \in X$ that minimizes a vector of objective functions

$$\mathbf{f}(\mathbf{x}) = \arg\max([f_1(\mathbf{x}^*), f_2(\mathbf{x}), f_3(\mathbf{x}), \dots, f_n(\mathbf{x})]) \dots\dots\dots (1)$$

with, $f_1(\mathbf{x}) \rightarrow$ cost related to inventory management; $f_2(\mathbf{x}) \rightarrow$ penalties for deviation from consumption schedules; $f_3(\mathbf{x}) \rightarrow$ losses due to inefficiencies in gas flow. $\in X$

2. Intent understanding and task decomposition

Let I represent the set of user intents, and T the set of tasks derived from these intents. The mapping from intents to tasks can be represented as a function

$$\phi: I \rightarrow T. \dots\dots\dots (2)$$

For a given intent $i \in I$, the task decomposition process can be formulated as:

$$T_i = \phi(i) \dots\dots\dots (3)$$

3. Form the swarm agent

Let A be the set of all agents, and C the set of distributed clusters formed by these agents. The process of agent cluster formation can be modelled as a function $\psi: A \rightarrow C$, which assigns agents to clusters based on predefined criteria such as role (e.g., flow management, consumption management) and scenario requirements.

$$C = \psi(A) \dots\dots\dots (4)$$

4. Determine self optimisation criterion for distribute clusters

Within each sub-cluster $c \in C$, let $A_c \subset A$ be the set of agents belonging to cluster c . The self-optimisation process within a sub-cluster can be represented as:

$$\mathbf{x}_c^* = \arg \mathbf{x}_c \in X_c \min_{j \in A_c} g_j(\mathbf{x}_c) \dots\dots\dots (5)$$

where $g_j(\mathbf{x}_c)$ represents the local objective function for agent j within cluster c .

5. Human-in-the-loop (HIL) interaction

Let H represent human interventions, and $\theta: H \times X \rightarrow X$ a function that incorporates human interactions into the decision-making process. The updated solution after human intervention can be expressed as:

$$\mathbf{x}' = \theta(h, \mathbf{x}) \dots\dots\dots (6)$$

where $h \in H$ is a specific human interventions input.

6. Model comparison or results merging

Let M be the set of models (or base models) under consideration. The comparison between two models $m_1, m_2 \in M$ can be represented by a comparison function $\kappa: M \times M \rightarrow R$, where:

$$\kappa(m_1, m_2) = \text{metric}(m_1, m_2) \dots\dots\dots (7)$$

if comparison process is triggered

The merging of models, either manually or autonomously, can be represented as a merging function

$$\mu: M \times M \rightarrow M \dots\dots\dots (8)$$

with

$$m' = \mu(m_1, m_2) \dots\dots\dots (9)$$

7. Side-chain thinking process

Let B represent the set of possible side-COT processes. The incorporation of a side-chain $b \in B$ into the main workflow can be modeled as:

$$\mathbf{x}'' = \text{incorporate}(b, \mathbf{x}') \dots \dots \dots (10)$$

where incorporate is a function that integrates the insights from the side-chain into the main decision making process $\mathbf{x}'$.

8. Template import and generation

Let T_{hist} and T_{ext} be the sets of historical and external templates, respectively. The import and generation process using generative agents can be represented as:

$$T_{\text{new}} = \text{generate}(T_{\text{hist}}, T_{\text{ext}}) \dots \dots \dots (11)$$

9. Multi-models Coordination

Let D be the set of models and algorithms (e.g., industrial software, other necessary AI models, etc) that can be coordinated. The coordination process can be formulated as:

$$r = \text{coordinate}(D, \mathbf{x}^*) \dots \dots \dots (12)$$

10. Output, recommendation and ranking

The final output, including predictions, simulations, insights, and recommended solutions, can be represented as a tuple:

$$\text{Output} = (\text{Prediction}, \text{Simulation}, \text{Insight}, \text{Top-N Recommendations}) = \mathbf{f}(\mathbf{x}) \dots \dots \dots (13)$$

where, $N \in \text{User determined and}$

$$\mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})] \rightarrow \text{argmax}\{\text{result} | T_i, C, \mathbf{x}_c^*, \mathbf{x}', \kappa, m', \mathbf{x}'', T_{\text{new}}, D\} \dots \dots \dots (14)$$

Application architecture of COTP

The system architecture and execution flow of COTP are illustrated in [Fig. 2](#). The system consists of a platform layer and an application layer. At the system's foundation, the foundation model management platform enables the integration of various foundation AIGC models include language, time - series, vision, multi-modalities understanding and classical machine learning, such as DeepSeek, GPT4o, TimeGPT, and LSTM/LightGBM/Xgboost, etc. It also supports scientific computing and knowledge engineering algorithms, such as graph - augmented retrieval (Graph-RAG) and multi-objective collaborative optimisation solvers (OR-OPS). The platform allows for the optional selection of foundation models for encompassing heterogeneous model integration to deal with complex tasks among CLGP operation. Additionally, the platform's connectivity function facilitates data integration from a variety of sources including PPS, SCADA, historical gas transmission volumes/plans, historical consumption volumes, turnover and balance management logs, and sensitivity factors, etc.

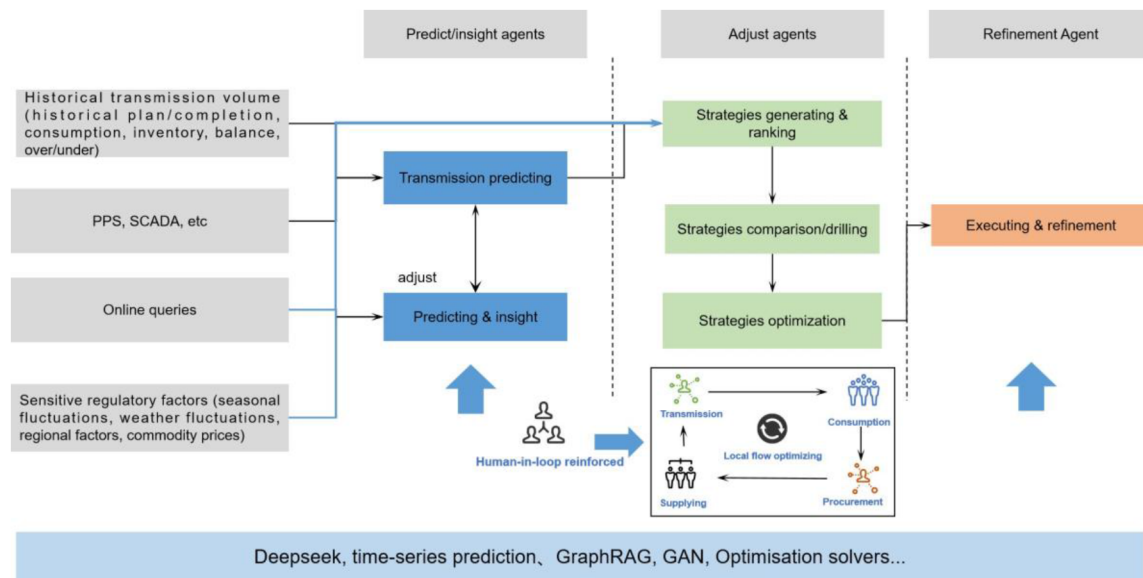


Figure 2—Technique roadmap for pipeline decision supporting based on human-in-loop

Upon application layer, a swarm agent structure (Chen et al. 2025) consists of prediction and insight agent (PIA), the adjustment factor agent (ADA), and the refinement agent (REA) is established to achieve CLGP coordinated scheduling as well as auxiliary decision-making optimisation. Within each agent, a distributed decision making mechanism (Chen et al. 2024) is constituted by one supervisory role agent, several specific role agents, and one feedback agent, respectively. The PIA executes thinking and task - flow generation processes. It schedules prediction models based on LSTM and time - series regression (LightGBM/Xgboost). Using historical CLGP transmission data as a basis to predict various combinations of gas transmission volumes for future time windows. By combining historical knowledge and CLGP operational mechanisms accumulated from pipeline operations, it provides insightful explanations for the prediction results. The combined "prediction + insight" results are then transmitted to the ADA. During the process of solving and transmitting the prediction and insight results, immediate intention intervention through human-in-the-loop (HIL) is supported to ensure the rationality of the prediction and insight outcomes.

Within the ADA, a distributed agent cluster composed of transmission volume, consumption volume, procurement volume, and supply volume (TCPS-Chain) agents (Chen et al., 2024), as well as a decision-making agent (DSA), is established. The DSA possesses autonomous feedback and execution capabilities. The distributed agent cluster performs local flow-optimising tasks (Chen et al., 2025; Zhang et al., 2025). By correlating macro/micro level data sources, the ADA receives the "prediction + explanation" results pushed by the PIA. Combining these results with the TCPS-Chain, COTP system generates CLGP scheduling strategies through a combination of an autonomous reflection mechanism designed with distributed swarm agents and in-depth thinking via AIGC. The DSA then conducts a preliminary ranking of these scheduling strategies. Given that the generated insight strategies are subject to multiple constraints and exhibit diverse characteristics, a side-chain thinking function is set up through the distributed agents. In order to maximise all potentials and to find even precise insight, the side-chain allows for reasoning in a bypass thinking chain in the form of HIL, targeting specific data, scenarios, and constraints without interfering with the main reasoning process. Reasonable reasoning results are enabled to incorporate into the main thinking chain to keep the thinking process through the right direction.

The scheduling policies and their ranking combinations generated by the ADA are pushed to the REA. The REA employs an attention spreading mechanism to achieve the adjustment and final refinement (Chen et al., 2025). By conducting online queries and integrating data from the CLGP scheduling expert

knowledge base, it makes decisions regarding the predicted and simulated gas transmission volumes as well as corresponding pipes operational conditions. This decision-making process takes into account the reverified prediction probabilities and in-depth thinking assisted by HIL. After decision-making, the execution workflow is generated by the REA and then validated through HIL. In conjunction with to-do lists, executable code is generated by agents and written into corresponding professional software or development environments via APIs or MCP (Krishnan, 2025), producing simulation results under multiple constraints. The supervisory role agent of the REA interacts with HIL to correct the generated results. If the REA is unable to correct the results autonomously, bad cases generated by both agents and users are fed back to the PIA, and the above- mentioned steps are repeated until satisfactory results are obtained.

Copanning and Coexecuting design of COTP

To dig in-depth the full potentials of generation and user-friendly decision-support capabilities of SGH, thereby achieving a higher level of human-machine friendliness, and offer more possibilities for precise decision making, the product frontend of COTP was designed. Through innovative human-machine interaction, it enables step-in/out based HIL human-machine collaborative analysis, thinking, planning, and decision-generating. The main interactive interface of COTP is illustrated in Fig. 3, where the interaction page adopts a step-by-step human-machine interaction mode to realise human-machine collaboration.

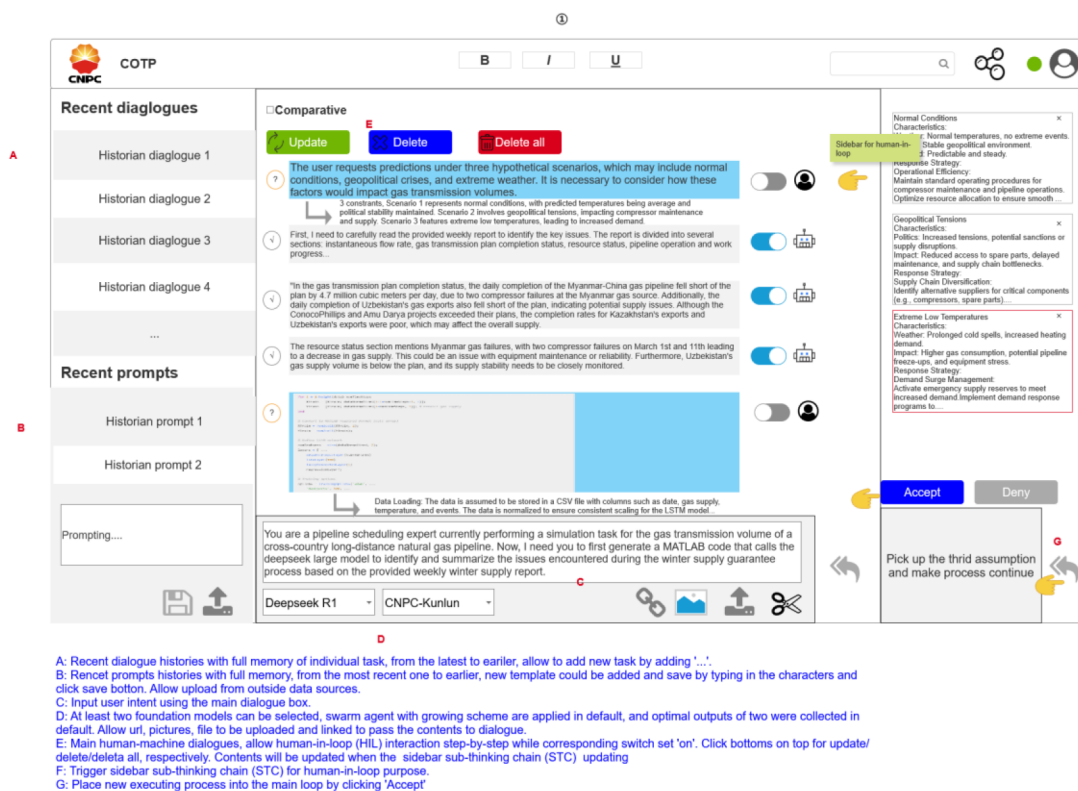


Figure 3—Interaction logics of COTP with comments

In response to the complex intentions of CLGP, the capabilities of issues finding, plan decomposition, distributed decision-supporting, autonomous execution, partial autonomous optimisation, and deep chain-of-thought provided by the swarm agents (DeepSeek. 2025, Schmidgall et al. 2025) are leveraged to execute the SGH workflow. This facilitates collaboration between agents and CLGP, enabling multi-pipeline complex simulations, interpretations, and decision-making assistance under multi-input and multiple constraint conditions.

Given the uncertain and fuzzy result characteristics of complex tasks in simulation and modelling, COTP incorporates toggle switches at each step of the thinking and executing processes. When each step of deep thinking is initiated, these switches can be used to set whether distributed autonomous decision-making by agents (Feng et al. 2025) is employed or whether HIL assisted thinking is triggered. During each decision drilling steps, if HIL thinking is triggered, the user can unfold a deeper level of reasoning (secondary reasoning) below the currently executing dialogue. This secondary reasoning is editable by the end-user. The user can supplement the auxiliary decision-making content through language and then continue with the decision process.

While triggering HIL interactions, to ensure that the main thinking chain remains undisturbed, a side chain interaction is designed using toggle switches at each step of thinking/execution and task decomposition. Through the side chain, a task chain based on other reasonable assumptions is operated whereby enabling users to establish experiment with other deep thinking and prompting modes in the side chain, to cover as many prediction and decision-making possibilities as necessary. When the results from the side chain meet the user's expectations, the user can, through the interactive interface provided by COTP, integrate the decisions and insights generated by the side chain into the corresponding positions of the main chain of thought and merge and execute the main chain of thought and task flow.

Through this design, multiple CLGP collaborative scheduling strategy simulations and emulations driven by accurate end-user demands are achieved across multiple countries, multiple pipelines, diverse policies and regulations, multiple languages, as well as differentiated scheduling strategies and standards. Effective control over multiple tasks is realised through innovative human-machine interaction and thought chains. Historical dialogues and prompts are retained on the left side in a reverse chronological order, from the most recent to the earliest. Users can resume the original simulation tasks by restarting historical dialogues based on the historical records tab. Meanwhile, the retained historical prompts support one-click pasting, further reducing the time spent on user input. In the upper right corner, insights can be shared within a specified tenant via links. Through the aforementioned design, more efficient simulation solutions and auxiliary decision-making are achieved. As the frontend interaction design illustrated in Fig. 3&4.

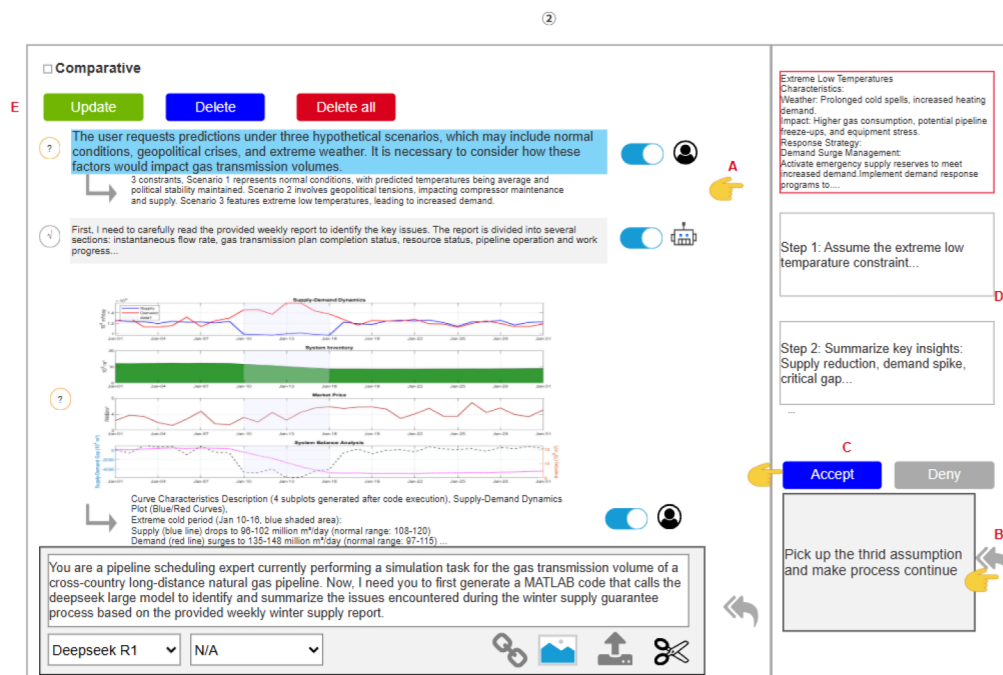


Figure 4—Predictive processing based on COTP and SGH with side chain embedded

To enable easier comparison, given that prediction, simulation, and decision support tasks are characterised by typical multi-constraint conditions and multiple possibilities, COTP has designed two simulation solution approaches in the frontend that support the comparison and complementary of dual models (Fig. 5) to facilitate decision support modelling and simulation. In the comparison mode, the product supports the generation of two distinct results and rankings through HIL based on different baseline models and pipeline scheduling algorithms. The end-user then selects one of the results to execute. Conversely, in the complementary mode, leveraging the main-side chain-of-thought fusion mode provided by SGH, the product autonomously integrates the decision support results generated by multiple models to form a final output for the end-user's reference. For both approaches, the output results require final decision-making by the end-user. Before executing the decision action, the end-user can edit and modify the textual content of the output results.

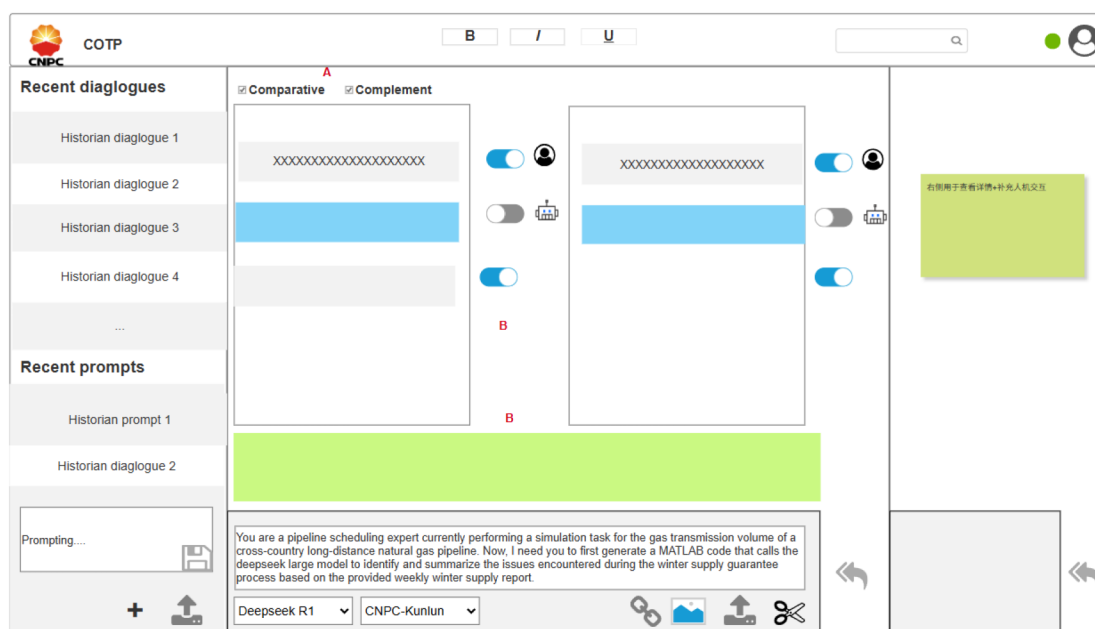


Figure 5—Interface interaction of comparative results with HIL upon side chain thinking

Field implementation of COTP in CLGP

The experimental study integrates 9 categories of operational data from 6 CLGP pipelines over the period from year 2018 to 2024, including CLGP historical scheduling reports, financial data, SCADA data, PPS data, inventory records, natural disaster information, geopolitical data, legal and regulatory frameworks, and policies of the countries where CLGP operates (segmented by region), to form a comprehensive integrated dataset. The document-based data comprises over 800 pieces, exceeding 2 million words (around 1.3 million tokens) while the time-series and tabular data exceed 180 million entries. COTP incorporates foundation models such as QWEN-13B/33B/72B, CNPC-KUNLUN-LLM 13B/33B/70B, and Deepseek R1/V3 to compare and evaluate predictive performance. The simulation system employs LSTM/LIGHTGBM/TFT/XGBOOST models as well as operational research solvers as tools for gas throughput simulation, while knowledge graphs and online query functionalities support the interpretation of predictive results. The AIGC foundation models are deployed by renting a HW Ascend 910B cluster, and the knowledge base and related data are deployed on a 32C/64G CentOS server cluster. For professional operational optimisation solver software, given the integration of decision-making and machine learning, collaboration with the CLGP system is achieved through MATLAB and Python IDEs. Communication between professional software applications is facilitated via API protocols and the MCP (Krishnan, 2025) protocol.

12 pipeline scheduling examiners (6 recruited online and 6 selected offline, 6 males and 6 females) were recruited to participate in the evaluation. As this study focuses on addressing HIL based decision support issues for CLGP, volunteers were instructed to input 50 abnormal events based on their past work experiences within their operational scope. COTP guided the volunteers in applying the COTP product by triggering abnormal events. Human-machine collaborative scheduling was initiated in response to events such as extreme weather conditions and abnormal gas throughput. Volunteers compared the evaluation results of directly using the foundation models (Baseline) and COTP across the aforementioned dimensions through a human-machine interaction interface. Both baseline and COTP interactions were conducted via natural language human-machine interaction, directly engaging with the foundation model and COTP interfaces, respectively. Natural language was used to drive the generation of gas throughput decisions and corresponding pipeline operation and maintenance strategies, as well as to provide explanations for the predictive and decision-making strategies. Volunteers conducted a combined subjective (by volunteer) and objective (similarity measured by base model) scoring evaluations, with each item scored on a scale of 0 to 5 (one decimal remaining). They performed qualitative and quantitative validations of the product's usability, user experience, predictive accuracy, and interpretability across multiple dimensions. Among these, professional decision support capabilities included issue finding, code generating, efficient output ranking, additional supports based on insight, and refinement suggesting will be established by using the agents.

Results and discussion

The experimental results presented in Fig. 6 and 7 demonstrate that, in terms of product user experience, all volunteers acknowledged the steerability, usability, and user-friendly of COTP, with scores significantly higher than those obtained when using baseline. Moreover, the innovative design of COTP in HIL and step-in interaction offers greater controllability and stability compared to directly employing the baseline. Regarding the issue findings, participants unanimously considered COTP's autonomous problem discovery capability and robustness to be significantly outperform to the baseline. This indicates that the step-in HIL interaction design and the loosely coupled bypass chain-of-thought approach contribute to a notable enhancement in user experience for problem sorting in complex decision-making scenarios within the CLGP.

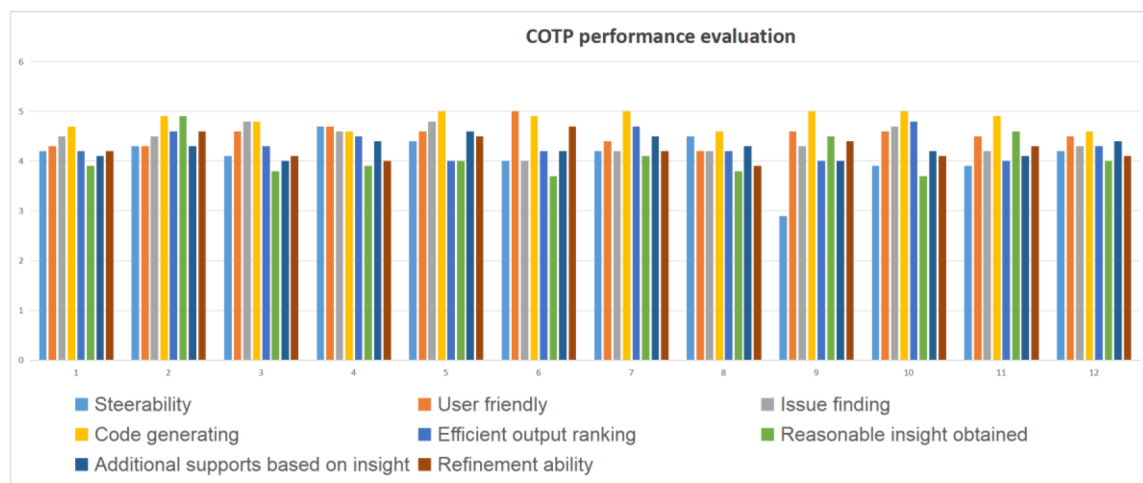


Figure 6—Experimental results--COTP performance evaluation

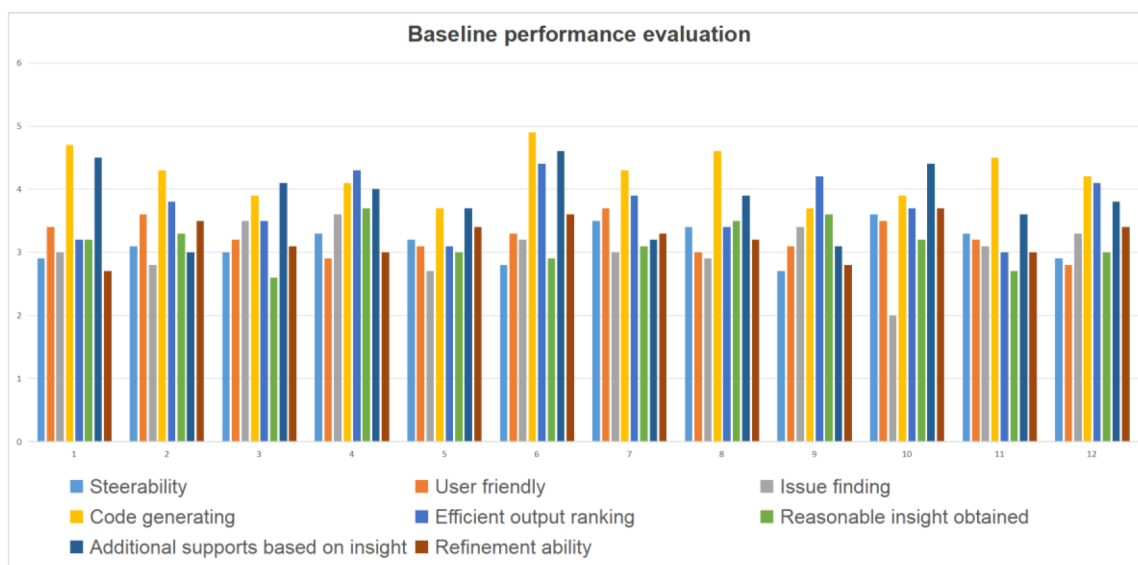


Figure 7—Experimental results--baseline performance evaluation

In the aspects of workflow orchestration and execution, encompassing effective task decomposition, workflow generation, code generation, efficient output ranking and execution, COTP can leverage a composite COT to achieve effective task assessment, generation of to-do lists, executable code generation, output result sorting, and hyper-automated execution of complex workflows, with the majority of users' scores distributed within the 4-5 point range. In terms of reasonable insight obtained, and addition supports generation, the insights generated through COTP's multi-round, step-in human-machine interaction, upon with side-chain supporting, exhibit greater interpretability, rationality, and usability compared to those directly generated by baseline, with the baseline's scores fluctuated among 2.7-4.6 across the three aforementioned indicators, whereas COTP demonstrated relatively stable performance, with its corresponding indicators fluctuating among 3.7-4.9. Regarding the system's autonomous optimisation capability (refinements), COTP significantly outperformed the direct adoption of baseline. In user ratings of this of this category, all scores for baseline were ≤ 4 , while those for COTP were consistently ≥ 4 . This suggests that COTP can effectively implement digital and intelligent predictions, reasonable data insights, provide valuable supplementary auxiliary decision-making information, and possess a good capacity for continuous optimisation when dealing with complex CLGP scheduling tasks. In contrast, the baseline exhibited significant fluctuations in performance across the same evaluation dimensions, producing a substantial proportion of ineffective information.

The aforementioned experimental results indicate that, for complex transnational pipeline networks composed of multiple CLGPs and the cross procedures complex decision-making chains formed based on these networks, which are characterised by intrinsic complexity, multiple heterogeneous constraints, and complex, time-varying uncertainties, the inherent chain-of-thought of foundation model struggles to traverse all decision-making prerequisites and provide more accurate decisions. Through the construction of comparative learning, a primary-bypass thinking chain, and step-by-step human-machine interaction in COTP, an imaginable, interpretable, convergent, and autonomously optimisable CLGP auxiliary decision-making tool and methodology are suitable for complex decision-making processes. A product demonstration extracted from the actual-using system, is illustrated in Fig. 8 in order to provide an even direct impression regarding to the performance and human-machine interaction changes provided by COTP.

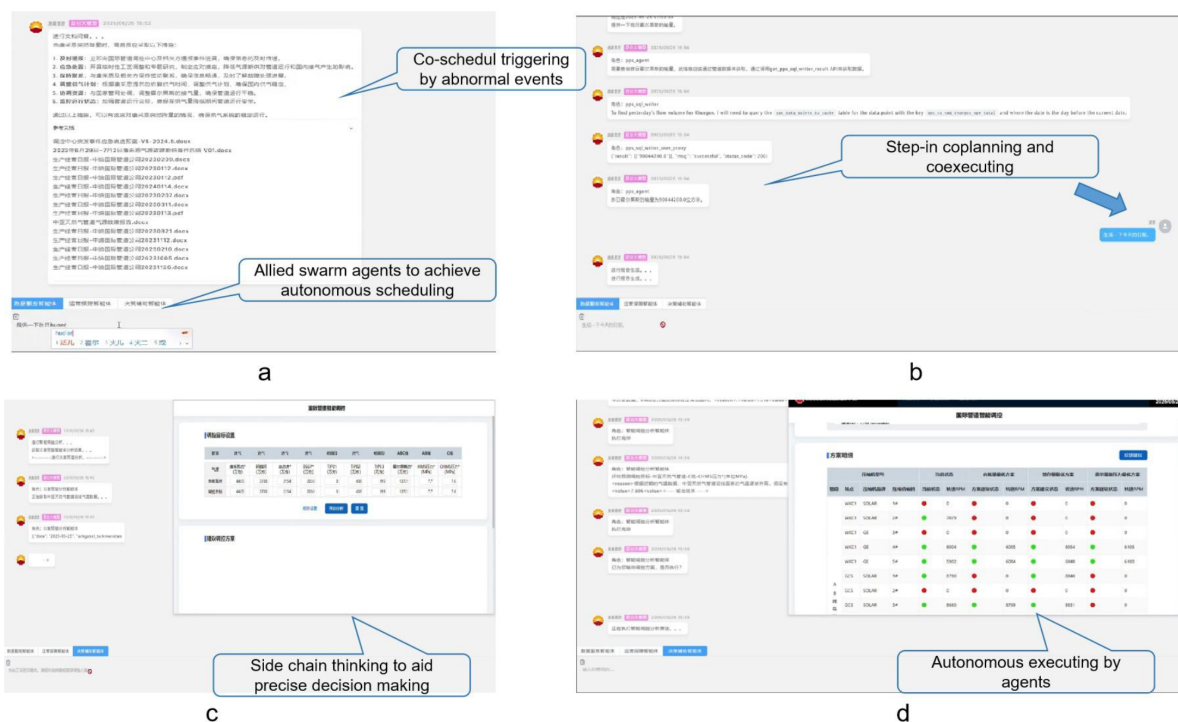


Figure 8—Demonstration of COTP embedded into CLGP for co-planning and co-executing, (a) Agents triggered by abnormal events; (b) Step-in coplanning and coexecuting with swarm agents and human in loop interaction (c) Side chain thinking taken to aid precise decision; (d) Autonomous executing by swarm agents

Concluding remarks

The manuscript proposed a swarm agent based co-planning and co-executing products, addressing the hyper-automation challenges of collaborative decision-making and execution for multiple CLGP transmission volume optimisation. By constructing a distributed agent cluster and orchestrating multi-role generative AI agents, the study develops a coplanning and coexecuting product to enable HIL based allied interaction to tackle fuzzy decision scenarios. This product and the corresponding human-machine interaction design achieves multi-objective collaborative optimisation of gas transmission volumes, equipment operational status, and operational policies simulation, thereby generating allied pipeline maintenance policies and transmission volume simulations for decision-supporting. To ensure feasible fault-tolerant control regarding to precise decision, a step-in HIL product interaction design is introduced to address uncertainties in future insights. The novel interaction interface provides stepwise intervention capabilities and contextual coherence to maintain correct reasoning chain direction, along with multi-model comparison functionalities that supply more on-demanding decision alternatives during co-planning and co-executing operations.

Validation across 6 CLGP operational practicing demonstrates that the COTP product significantly outperforms traditional methods in terms of steerability, user-friendliness, issue finding, code generation, insights ranking, and decision-making rationality across autonomous optimisation loops. These improvements underscore the efficacy of integrating distributed agent clusters with HIL based human-machine interaction, particularly for solving multi-objectives, multi-factor coordination problems under transnational, multi-pipeline, multi-workflow constraints encompassing macro/micro-scale operational boundaries. The implementation enables autonomous triggering, dynamic recombination, and continuous learning pathways for business-finance-technique integrated collaborative decision making and execution.

Integration with existing enterprise scheduling information systems facilitates a paradigm shift from planned scheduling to event driven decision-making, enabling users to discover even feasible and business-benefited scheduling policies. Leveraging COTP's multi-layered and step-in deep reasoning capabilities,

the product achieves substantial gains in user engagement while driving a transition to subscription based information system business models. This technical pathway demonstrates the utility of generative AI, agents, and pipeline operational scheduling technologies to synergistically address complex coordination challenges in cross-border energy infrastructure management with optimism business-finance balance.

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